Which note should ScentSelf use as its first test anchor?

Palaloy | MADT7202 | Update4 | Code, notebook & results: Pornphrom/MADT7202_midterm

1 The decision

Palaloy, ScentSelf founder, will select a test anchor. A stable bridge that outranks prevalence can support differentiation; otherwise use only a provisional control, with no positioning claim.

2 Why a network

Counts show common notes; weighted shortest paths show which notes connect note groups. Comparing betweenness and prevalence ranks distinguishes these roles.

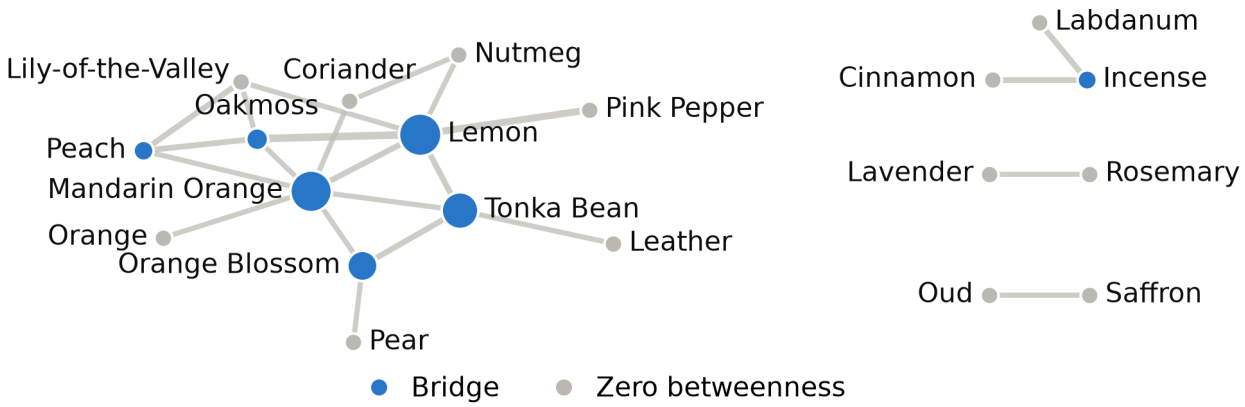
3 Data

Sources	Fragrantica; Parfumo note supplements in **11/209**. Selenium; offline note parsing.
Period	30 Aug-12 Oct 2026 (author-confirmed).
Volume	**209** perfumes, 356 notes; wider set: **1,780**. URL manifest in repo.
Access	Permission not established. Fragrantica ToS 5.3 restricts unapproved automation. Raw data not shared; source requirement unresolved.
Sampling	Purposive list; Floral 61 vs Leather 4. Not a market sample; supplements are not triangulation.

4 Graph construction

Nodes / edges	Cleaned notes / notes listed in the same perfume. No seed-note filter.
Weight	Count favours repeated pairings; distance = 1/count. NPMI >0.1 checks prevalence.
Pruning	Exclude prevalence >=20% (10 notes); retain frequency >=5 (89 notes). Drop edges <6 and isolates.
Choice	k=6: nearest density 0.12 among k=3..10; **20 nodes / 23 edges**. k=8 leaves 3 nodes / 2 edges.
Rule	Average ranks on same nodes; gap >=5, positive top-10, >=80% of 300 whole-brand resamples; top-10 at k=3,6,8.

5 The network



Node size = betweenness; blue = positive betweenness; line width = pair count.

6 Findings

Note	Freq. rank	BC rank	Gap	Top-10 rate
Lemon	1.5	1	0.5	87.3%
Peach	12.5	6	6.5	25.3%

1. Peach has a 6.5-rank advantage but only 25.3% stability. Lemon is stable (87.3%) with little rank advantage. Neither qualifies.

Check	Pass	Control
Raw (23 edges)	0	Lemon
NPMI (22 edges)	0	Lemon
No supplements (198)	0	Lemon
Wider (1,780)	0	Orange Blossom

2. Zero anchors pass all criteria in each check; the wider catalogue changes the control. Keep Lemon specific to the curated list.

7 Recommended action (proposed)

13 Oct - Palaloy: resolve source permission or select permitted replacement data and rerun before committing to the test.

14 Oct, if cleared: brief matched-base prototypes: Lemon control vs Lemon + Peach; hold concentration constant.

15 Oct, if ready: run a blinded, order-balanced pilot with 20 testers. Advance Peach only if at least 15 prefer it; otherwise retain the control. Postpone if prerequisites fail.

8 What would change my mind

Declared-note co-occurrence is not liking or a recipe. The graph changes with cutoff and catalogue. A permitted, broader sample that changes the control, or failure in the blinded pilot, would change the recommendation.